RR-D-640B
November 19, 1975
SUPERSEDING
Fed. Spec. RR-D-640A
November 18, 1968

FEDERAL SPECIFICATION

DRINKING FOUNTAINS, PORTABLE

(PRESSURE AND GRAVITY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable drinking fountains.

1.2 Classification.

1.2.1 Types. The portable drinking fountain shall be of the following types as specified (see 6.2):

Type I - Pressure (bubbler).
Type II - Gravity (individual cup).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-390	- Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium, and High Density).
QQ-A-250/2	- Aluminum Alloy 3003, Plate and Sheet.

FSC 4510

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QQ-S-763	- Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.
QQ-S-766	- Steel Plates, Sheets and Strip - Corrosion Resisting.
UU-C-806	- Cup, Paper, Lids, Paper or Lids Plastic, Disposable.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-D-2917	- Dispensers, Salt Tablet.

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Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The portable drinking fountain shall consist of an outer shell cover, inner liner, insulation, and facilities for hanging or carrying.

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3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Materials intended for use with portable drinking fountains shall impart no taste or odor to the water. Materials shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procurement activity to the appropriate department medical service who will act as an advisor to the procuring agency (see 6.3).

3.2.1 Steel, corrosion-resisting. Corrosion-resisting steel shall conform to QQ-S-763, or QQ-S-766, Class 302, 303, 304, or 305.

3.2.2 Aluminum sheet 3003. Aluminum sheet 3003 shall conform to QQ-A-250/2, Temper H12.

3.2.3 Plastic. Plastic material specified herein shall be virgin material; however, reground imperfect parts, sprues and runners due to molding or forming imperfections of the same composition as the virgin material shall be accepted.

3.2.3.1 Plastic molding material. Polyethylene (low and medium density) shall conform to L-P-390, Type I, Class H, Grade 2 or 3.

3.3 Outer shell. The outer shell shall be made of corrosion-resisting steel, galvanized steel or aluminum, as specified (see 6.2). The wall of the shell shall be of a thickness, that when filled and hanging from a support bracket, no sign of deforming shall be in evidence. The wall shall extend beyond the bottom surface to afford stability when standing on the bottom.

3.4 Inner liner. The inner liner shall be of corrosion-resisting steel, aluminum alloy or plastic, as specified (see 6.2). If of metal, the liner wall shall be formed from one piece, rolled into shape, and shall have a watertight seal. The bottom shall be of one piece, and attached to the wall of the liner with a watertight seal. When the inner liner is of plastic, it shall be molded into one piece.

3.5 Insulation material. The insulation material shall be packed between the inner liner and outer shell, and shall be of a nonhygroscopic material which will not cause corrosion or in any way harmfully affect the container. The insulation material shall be of a type that the liquid temperature rise after 8 hours shall be not more than 20° F when tested in accordance with 4.4.2.1.

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3.6 Capacity. Unless otherwise specified (see 6.2), the drinking fountain shall have a capacity of 4 gallons of liquid.

3.7 Brackets and carrying devices.

3.7.1 Brackets. Unless otherwise specified (see 6.2), the drinking fountain shall be equipped with brackets for supporting the fountain when hung from a wall or the body of a vehicle. The brackets shall be either welded or riveted to the fountain. The bracket that mounts on the wall or vehicle body shall be complete with mounting screws. The brackets shall support the drinking fountain when filled with water without evidence of deformation or failure, when tested in accordance with 4.4.2.2.

3.7.2 Carrying handles. Unless otherwise specified (see 6.2), the fountain shall be equipped with carrying handles. The handles shall be mounted to the shell and located diametrically opposite each other. There shall be no sharp edges, the handles shall afford a comfortable handgrip and conform to the applicable requirements of MIL-STD-1472.

3.7.3 Carrying strap. When specified (see 6.2), the fountain shall be equipped with an adjustable cotton webbing shoulder strap, not less than 1-1/2 inches in width, conforming to MIL-W-530, Type IIa, IIb or III, Class 1b. The strap shall be provided with an adjustable rustproof buckle to enable the carrier to adjust the strap to size, and fittings, for attachment to the tank.

3.8 Salt tablet dispenser. When specified (see 6.2), a salt tablet dispenser conforming to MIL-D-2917 shall be furnished with the fountain, except that the capacity shall be 500 tablets. A bracket shall be attached to the fountain for holding the dispenser.

3.9 Joints or seams. The joints or seams shall be welded, cemented, or double locked and soldered. The joints or seams shall not leak when tested in accordance with 4.4.2.3.

3.10 Treatment and painting. Unless otherwise specified (see 6.2), the outside and cover of the fountain shall be treated and painted in accordance with MIL-T-704, Type B. The inside of the fountain shall be cleaned.

3.11 Identification marking. Each fountain shall be marked on the bottom or side in accordance with MIL-STD-130.

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3.12 Type I pressure fountain. The Type I fountain shall be equipped with a drinking nozzle and a press type control valve for control of the liquid flow. The nozzle shall be of corrosion-resisting material and shall be at an angle so as to prevent the return of liquid onto the nozzle while being used. The nozzle shall be accessible for cleaning. The fountain shall be equipped with a hand pump to supply pressure for the discharge of the water.

3.12.1 Guard. The nozzle shall be protected by a guard designed in conjunction with the nozzle to prevent persons from directly contacting the nozzle while drinking, and to prevent foreign matter from dropping vertically onto the nozzle. Spaces between nozzle and guard shall allow convenient cleaning of those parts. The guard shall be of corrosion-resisting material, or shall be nickel- or chrome-plated.

3.12.2 Valve. The valve that controls the flow of water shall be conveniently located and shall be of the quick-operating type in order to prevent waste. The valve shall operate from a push-type control mounted on the fountain.

3.12.3 Water feed tube. The water shall be fed to the bubbler head by a tube having an intake located near the bottom of the container. A removable perforated strainer shall be attached to the lower end of the tube. The tube shall be attached to the lower end of the bubbler head in such a manner that it may be removed for cleaning.

3.12.4 Cover. The cover, when closed and locked into position, shall provide an airtight, leakproof seal, and shall be equipped with a flexible type one-piece replaceable rubber gasket.

3.12.5 Hand pump. The hand pump shall develop sufficient pressure to force the water out of the nozzle. The fountain and piping shall be constructed, using necessary valves, so as to provide a uniform stream of water 4 inches high, plus or minus 1/2 inch, from the nozzle after the buildup of the pressure. The pump shall be designed to prevent lubricants from leaking into the water, and to minimize the possibility of injury to the fingers. The hand pump shall have a safety valve to prevent excess pressure.

3.12.6 Spill cup. When specified (see 6.2), the bubbler type fountain shall be equipped with a spill cup. The cup shall be of corrosion-resisting material. The cup shall have a minimum 2-quart capacity, and shall be mounted to the fountain by a bracket.

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3.13 Type II gravity fountains. The Type II individual cup drinking fountain shall be of gravity feed construction. A removable, fitted cover, with a handle, shall be provided to protect the water in the fountain. A vent shall be provided in the cover or in the body to permit the entrance of air into the reservoir as water is used.

3.13.1 Faucet. The faucet shall be mounted on the side and near the bottom of the fountain, and shall be of the replaceable type. The faucet shall be corrosion-resisting steel, aluminum or plastic, as specified (see 6.2). The faucet shall be the quick-acting, self-closing type.

3.13.2 Used cup receptacle. When specified (see 6.2), the fountain shall be provided with a used cup receptacle. The receptacle shall be so constructed as to prevent it being mistaken for a cup dispenser. The receptacle and fountain shall be equipped with mounting brackets.

3.13.3 Cup dispenser. When specified (see 6.2), the drinking fountain shall be equipped with a cup dispenser of tube-type design, with cups being loaded at the top and extracted from the bottom. The cups shall be of 3-1/2-ounce capacity conforming to UU-C-806, Type I, Style B. The dispenser and fountain shall be equipped with a bracket for hanging the cup dispenser.

3.14 Workmanship. The fountain shall be clean, and all joints shall be free of any defect which may adversely affect serviceability or may impede cleaning of the fountain. All seams shall be sanded as smooth as the parent surface. There shall be no burrs or slivers that would cause injury when washing or carrying the fountain.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. All fountains of the same type offered for delivery at one time shall constitute a lot.

4.3.2 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.4 Test. Samples selected in accordance with 4.3.2 shall be tested as specified in 4.4.2. AQL shall be 1.00 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The fountains shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials in contact with potable water not as specified or not approved.
103. Components missing or not as specified.
104. Capacity not as specified.
105. Brackets and carrying devices not as specified.

Minor

201. Treatment and painting not as specified.
202. Marking not as specified.
203. Workmanship not as specified.

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4.4.2 Tests.

4.4.2.1 Temperature rise. The fountain shall be subjected to room temperature (70° F, plus or minus 2° F) for at least 4 hours. After the 4 hours, 2 gallons of a liquid at a temperature of 50° F, plus or minus 2° F, shall be added. The fountain, with cover in place, shall then be placed in a room with the temperature controlled at 90° F, plus or minus 2° F. At the end of 8 hours, the gain in temperature of the liquid shall be measured. A temperature rise of more than 20° F shall constitute failure of this test.

4.4.2.2 Bracket. A fountain filled with water shall be hung by the support bracket. Any visible give, protruding, sagging, or deforming of brackets or shell shall constitute failure of this test.

4.4.2.3 Leaking seam. The complete fountain shall be tested for leaking seams by submerging a fountain, less cover, in water. Any evidence (air bubbles) of the liner or shell leaking water into the insulation area shall constitute failure of this test.

4.4.2.4 Pressure. The Type I fountain shall be filled with water and pumped to normal pressure with a hand pump. The nozzle shall be opened to discharge a stream of water. A stream of water less than 4 inches, plus or minus 1/2 inch, shall constitute failure of this test. The pressure shall be increased with the hand pump until the safety valve pops off. Failure of the safety valve to pop off before rupture of the fountain shall constitute failure of this test. Any evidence of lubricant leaking into the water shall also constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

106.	Materials and containers not as specified for Level A or B. Each incorrect material or container shall constitute one defect.
107.	Removable and loose components not packaged as specified for Level A.

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108. Contents not cushioned as specified for Level A.
109. Box not closed and sealed as specified for Level A.
110. Gross weight or size exceeds the box limitations for Level A or B.
111. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. Each complete fountain shall be packaged in a close-fitting box conforming to PPP-B-636, W5c, style optional. Prior to placing the fountain in the box, all removable and loose components shall be packaged in heat-sealed plastic bags. The bags shall be placed within the fountain or the box as space permits. All contents shall be cushioned and secured in a manner to prevent movement or damage. The box shall be closed and sealed as specified for Method V in the appendix to the box specification.

5.1.2 Level C. Each complete fountain shall be packaged to provide protection against mechanical or physical damage from the supplier to the initial destination. The supplier's standard commercial packaging is acceptable providing it fulfills these requirements.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Fountains, packaged as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional, in quantities not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Fountains, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-640, Class 2, style optional, or PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

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5.2.3 Level C. Fountains, packaged as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in wooden or fiberboard containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with FED. STD. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Portable drinking fountains are intended for carrying water to workers not positioned near a stationary source of water. The water is either precooled or is cooled with ice loaded into the tank. The used cup receptacle and spill cup are generally for indoor use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of fountain required (see 1.2.1).
(c) Material required (see 3.3, 3.4, 3.13.1).
(d) Capacity if other than as specified (see 3.6).
(e) When brackets are not required (see 3.7.1).
(f) When carrying handles are not required (see 3.7.2).
(g) When straps are required (see 3.7.3).
(h) When a salt tablet dispenser is required (see 3.8).
(i) When treatment and painting other than as specified is required (see 3.10).
(j) When a spill cup is required (see 3.12.6).
(k) When a used cup receptacle is required (see 3.13.2).
(l) When a cup dispenser is required (see 3.13.3).
(m) Level of packaging and level of packing required (see 5.1 and 5.2).

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6.3 The Surgeon General will act as advisor to make the necessary determinations on health hazards.

MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD
Air Force - 84

Review Activities:

Army - MD
DSA - CS

User Activity:

Navy - MC

Civil Agency Coordinating Activities:

GSA-FSS
DC-DCG
HEW-FEC

Project No. 4510-0109

Preparing Activity:

Army - ME

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DOCUMENT IDENTIFIER AND TITLE

RR-D-640B Drinking Fountains, Portable (Pressure and Gravity)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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